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328873(28)

B. E. (Eighth Semester) EXAMINATION, Nov.-Dec., 2014

(Branch : Et & T)

INFORMATION THEORY AND CODING

(PE—III)

Time : Three Hours] [*Maximum Marks : 80*

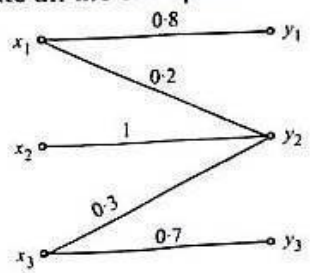
[*Minimum Pass Marks : 28*

Note : The questions paper contains *five* questions in all. All questions are compulsory. Part (a) of each question is compulsory. Attempt any *two* parts from the rest.

Unit—I

1. (a) What do you mean by information theory ? 2

(b) A discrete source transmits messages x_1, x_2, x_3 with probabilities 0.3, 0.4, 0.3. The source is connected to the channel given in figure below. Calculate all the entropies. 7



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(c) What is Channel Capacity ? A Gaussian channel has 1 MHz bandwidth. Calculate the channel capacity if the signal power to noise spectral density ratio is 10^5 Hz (5/η). Find the information rate. 7

(d) Show that : 7

$$H(X, Y) = H(X/Y) + H(Y)$$

Unit—II

2. (a) Define code word and code rate. 2

(b) An analog signal band limited to 5 kHz is quantized in 8 levels of a PCM system with probabilities of $\frac{1}{4}, \frac{1}{5}, \frac{1}{5}, \frac{1}{10}, \frac{1}{10}, \frac{1}{20}, \frac{1}{20}$ and $\frac{1}{20}$ respectively. Find the entropy and the rate of information. 7

(c) The parity check matrix $\bar{H}$ of a linear (7, 4) block code is given as the follows. Show how data words (i) 0011 (ii) 0100 and (iii) 0101 are coded. 7

$$\bar{H} = \begin{pmatrix} 1 & 0 & 1 & 1 & 1 & 0 & 0 \\ 1 & 1 & 0 & 1 & 0 & 1 & 0 \\ 0 & 1 & 1 & 1 & 0 & 0 & 1 \end{pmatrix}$$

(d) Explain linear tree code by matrices, with an example. 7

Unit—III

3. (a) What do you mean by Compression ? 2

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(b) What is the difference between lossless and lossy compression ? What are different steps for image compression ? 7

(c) What is the difference between static and dynamic Huffman coding ? Explain with basic mechanism of Huffman coding. 7

(d) Write short notes on the following : 7

(i) CC/TT group 3 1D compression !

(ii) CC/TT group 3 2D compression.

Unit—IV

(a) Briefly state the requirements of full motion video compression. 2

(b) Explain CC/TT H 261 video coding algorithm. 7

(c) Explain MPEG-2 with the block diagram of its coder and decoder. 7

(d) What are the types of Audio Compression ? 7

Unit—V

(a) What is the difference between conventional cryptography and public key cryptography ? 2

(b) Explain DES in brief. 7

(c) Explain Diffe-Hellman public key distribution with an example. 7

(d) Explain RSA algorithm with an example. 7

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